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Non-contact seal for rotational equipment with radial through-hole

Abstract

An assembly is provided for rotational equipment. This assembly includes a plurality of seal shoes, a seal base and a plurality of spring elements. The seal shoes are arranged about a centerline in an annular array. The seal shoes include a first seal shoe configured with an aperture that extends radially through the first seal shoe. The seal base circumscribes the annular array of seal shoes. The spring elements include a first spring element. The first spring element includes a first mount, a second mount and a spring beam. The first mount is connected to the first seal shoe. The second mount is connected to the seal base. The spring beam extends laterally between and is connected to the first mount and the second mount.

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Background/Summary

BACKGROUND OF THE DISCLOSURE

1. Technical Field

(1) This disclosure relates generally to rotational equipment and, more particularly, to a non-contact seal for rotational equipment.

2. Background Information

(2) Rotational equipment typically includes one or more seal assemblies for sealing gaps between rotors and stators. A typical seal assembly includes a contact seal with a seal element such as a knife edge seal that engages a seal land. Such a contact seal can generate a significant quantity of heat that can reduce efficiency of the rotational equipment as well as subject other components of the rotational equipment to high temperatures and internal stresses. To accommodate these high temperatures and stresses, certain components of the rotational equipment may be constructed from specialty high temperature materials. However, these materials can significantly increase manufacturing and servicing costs as well as mass of the rotational equipment. While non-contact seals have been developed in an effort to reduce heat within rotational equipment, there is still room for improvement to provide an improved non-contact seal. In particular, there is room in the

art for a non-contact seal with improved damping characteristics.

SUMMARY OF THE DISCLOSURE

(3) According to an aspect of the present disclosure, an assembly is provided for rotational equipment. This assembly includes a plurality of seal shoes, a seal base and a plurality of spring elements. The seal shoes is arranged about a centerline in an annular array. The seal shoes include a first seal shoe configured with an aperture that extends radially through the first seal shoe. The seal base circumscribes the annular array of seal shoes. The spring elements include a first spring element. The first spring element includes a first mount, a second mount and a spring beam. The first mount is connected to the first seal shoe. The second mount is connected to the seal base. The spring beam extends laterally between and is connected to the first mount and the second mount.

(4) According to another aspect of the present disclosure, another assembly is provided for rotational equipment. This assembly includes a plurality of seal shoes, a seal base and a plurality of spring elements. The seal shoes are arranged around a centerline and include a first seal shoe. The first seal shoe extends axially along the centerline between a seal shoe first side and a seal shoe second side. The first seal shoe includes a plurality of projections and an aperture. The projections are arranged into a grouping which is asymmetrically arranged axially along the centerline between the seal shoe first side and the seal shoe second side. The aperture extends radially through the first seal shoe. The aperture is located axially between the grouping and the seal shoe second side. The seal base circumscribes the seal shoes. The spring elements include a first spring element which is connected to and extends between the first seal shoe and the seal base.

(5) According to still another aspect of the present disclosure, still another assembly is provided for rotational equipment. This assembly includes a plurality of seal shoes, a seal base and a plurality of spring elements. The seal shoes are arranged about a centerline and include a first seal shoe. The first seal shoe includes a base, a plurality of projections and an aperture. The base extends axially along the centerline between a first side of the first seal shoe and a second side of the first seal shoe. Each of the projections extends radially inward from the base. The projections include a first projection and a second projection. The first projection and the second projection are located axially between the first side of the first seal shoe and an axial midpoint of the first seal shoe. The aperture extends radially through the base. The aperture is located axially between the projections and the second side of the first seal shoe. The seal base circumscribes the seal shoes. The spring elements include a first spring element that is connected to and extends between the first seal shoe and the seal base.

(6) The aperture may extend axially within the first seal shoe.

(7) The aperture may extend axially into the first seal shoe.

(8) The aperture may extend laterally within the first seal shoe.

(9) The aperture may extend laterally into the first seal shoe.

(10) The aperture may be located laterally between the first mount and the second mount.

(11) The first mount may be located laterally between the second mount and the aperture.

(12) The aperture may be configured with an elongated cross-sectional geometry.

(13) The aperture may be configured with a circumferentially symmetric cross-sectional geometry.

(14) The aperture may be configured with a circular cross-sectional geometry.

(15) The first seal shoe may also be configured with a second aperture that extends radially through the first seal shoe.

(16) The first seal shoe may include a base and a plurality of projections. The base may extend axially along the centerline between a first side of the first seal shoe and a second side of the first seal shoe. Each of the projections extends radially inward from the base. The aperture may extend radially through the base and/or may be located axially between the second side of the first seal shoe and the projections.

(17) The projections may include a first projection and a second projection. The first projection and the second projection may be located axially between the first side of the first seal shoe and an

axial center of the first seal shoe.

(18) The projections may also include a third projection. The third projection may be further located axially between the first side of the first seal shoe and the axial center of the first seal shoe.

(19) The first seal shoe may extend laterally between a first end of the first seal shoe and a second end of the first seal shoe for a seal shoe length. The aperture may extend laterally for an aperture length which is between fifty percent and eighty percent of the seal shoe length.

(20) The first seal shoe may extend laterally between a first end of the first seal shoe and a second end of the first seal shoe for a seal shoe length. The aperture may extend laterally for an aperture length which is between ten percent and fifty percent of the seal shoe length.

(21) The first seal shoe may extend laterally between a first end of the first seal shoe and a second end of the first seal shoe for a seal shoe length. The aperture may extend laterally for an aperture length which is between one percent and ten percent of the seal shoe length.

(22) The first spring element may also include a second spring beam that extends laterally between and is connected to the first mount and the second mount.

(23) The assembly may also include a stationary structure, a rotating structure and a seal assembly. The rotating structure may be configured to rotate about the centerline. The seal assembly may include the seal shoes, the seal base and the spring elements. The seal assembly may be configured to seal a gap between the stationary structure and the rotating structure. The seal shoes may be arranged circumferentially about and may sealingly engage the rotating structure. The seal base may be mounted to the stationary structure.

(24) The present disclosure may include any one or more of the individual features disclosed above and/or below alone or in any combination thereof.

(25) The foregoing features and the operation of the invention will become more apparent in light of the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a partial side sectional illustration of an assembly for rotational equipment.

(2) FIG. 2 is a partial side sectional illustration of the assembly configured with an alternative stationary structure.

(3) FIG. 3 is a perspective general schematic illustration of a primary seal device for the assembly.

(4) FIG. 4 is a partial side sectional illustration of the primary seal device.

(5) FIG. 5 is a cross-sectional illustration of a portion of the primary seal device.

(6) FIG. 6 is a partial side sectional illustration of the primary seal device configured with alternative projections.

(7) FIGS. 7A and 7B are perspective illustrations of a seal shoe for the primary seal device configured with an aperture.

(8) FIG. 8 is a perspective illustration of the seal shoe configured with an alternative aperture.

(9) FIG. 9 is a perspective illustration of the seal shoe configured with another alternative aperture.

(10) FIG. 10 is a perspective illustration of the seal shoe configured with still another alternative aperture.

(11) FIG. 11 is a cross-sectional illustration of a portion of the primary seal device configured with the seal shoe of FIG. 10.

(12) FIGS. 12A and 12B are perspective illustrations of the seal shoe configured with a plurality of apertures.

(13) FIGS. 13A and 13B are perspective illustrations of the seal shoe configured with alternative apertures.

(14) FIG. 14 is a perspective illustration of the seal shoe configured with other alternative

apertures.

(15) FIG. **15** is a perspective illustration of the seal shoe configured with other alternative apertures.

(16) FIG. **16** is a perspective illustration of the seal shoe configured with other alternative apertures.

(17) FIG. **17** is a perspective illustration of the seal shoe configured with still other alternative apertures.

(18) FIG. **18** is a schematic illustration of an elongated aperture for the seal shoe.

(19) FIG. **19** is a schematic illustration of a circular aperture for the seal shoe.

(20) FIG. **20** is a schematic illustration representing lengths of an aperture and a seal shoe.

(21) FIG. **21** is a side cutaway illustration of a gas turbine engine which may be configured with the rotational equipment assembly.

DETAILED DESCRIPTION

(22) FIG. **1** illustrates an assembly **30** for rotational equipment with an axial centerline **32**, which centerline **32** may also be an axis of rotation (e.g., a rotational axis) for one or more components of the rotational equipment assembly **30**. An example of such rotational equipment is a gas turbine engine for an aircraft propulsion system, an exemplary embodiment of which is described below in further detail (e.g., see FIG. **21**). However, the rotational equipment assembly **30** of the present disclosure is not limited to such an aircraft or gas turbine engine application. The rotational equipment assembly **30**, for example, may alternatively be configured with rotational equipment such as an industrial gas turbine engine, a wind turbine, a water turbine or any other apparatus in which a seal is provided between a stationary structure and a rotating structure; e.g., a rotor.

(23) The assembly **30** of FIG. **1** includes a stationary structure **34**, a rotating structure **36** and a seal assembly **38** such as, for example, an adaptable non-contact seal assembly. The seal assembly **38** is mounted with the stationary structure **34** and configured to substantially seal an annular gap **40** between the stationary structure **34** and the rotating structure **36** as described below in further detail.

(24) The stationary structure **34** includes a seal carrier **42**. This seal carrier **42** may be a discrete, unitary annular body. Alternatively, the seal carrier **42** may be configured with another component/portion of the stationary structure **34**. The seal carrier **42** has a seal carrier inner surface **44**. This seal carrier inner surface **44** may be substantially cylindrical. The seal carrier inner surface **44** extends circumferentially about (e.g., completely around) and faces towards the axial centerline **32**. The seal carrier inner surface **44** at least partially forms a bore in the stationary structure **34**. This bore is sized to receive the seal assembly **38**, which may be fixedly attached to the seal carrier **42** by, for example, a press fit connection between the seal assembly **38** and the seal carrier inner surface **44**. The seal assembly **38**, of course, may also or alternatively be fixedly attached to the seal carrier **42** using one or more other techniques/devices.

(25) The rotating structure **36** includes a seal land **46**. This seal land **46** may be a discrete, unitary annular body. For example, the seal land **46** may be mounted to a shaft of the rotating structure **36**. Alternatively, the seal land **46** may be configured with another component/portion of the rotating structure **36**. For example, the seal land **46** may be an integral part of a shaft of the rotating structure **36**, or another component mounted to the shaft.

(26) The seal land **46** of FIG. **1** has an outer seal land surface **48**. This outer seal land surface **48** may be substantially cylindrical. The outer seal land surface **48** extends circumferentially about (e.g., completely around) and faces away from the axial centerline **32**. The outer seal land surface **48** is configured to face towards and is axially aligned with the seal carrier inner surface **44**. While FIG. **1** illustrates the outer seal land surface **48** and the seal carrier inner surface **44** with approximately equal axial lengths along the axial centerline **32**, the outer seal land surface **48** may alternatively be longer or shorter than the seal carrier inner surface **44** in other embodiments.

(27) The seal assembly **38** includes a primary seal device **50** and one or more secondary seal

devices **52**. The seal assembly **38** also includes one or more additional components for positioning, supporting and/or mounting one or more of the seal devices with the stationary structure **34**. The seal assembly **38** of FIG. **1**, for example, includes a first ring structure **54** configured for positioning, supporting and/or mounting the secondary seal devices **52** relative to the primary seal device **50**. This first ring structure **54** may also be configured for axially positioning and/or supporting an axial first side surface **56** of the primary seal device **50** relative to the stationary structure **34**. The seal assembly **38** of FIG. **1** also includes a second ring structure **58** (e.g., a scalloped support ring/plate) configured for axially positioning and/or supporting an axial second side surface **60** of the primary seal device **50** relative to the stationary structure **34**. However, the second ring structure **58** may be omitted where, for example, the second side surface **60** of the primary seal device **50** is abutted against another component/portion of the stationary structure **34** (e.g., an annular or castellated shoulder) or otherwise axially positioned/secured with the stationary structure **34**; e.g., see FIG. **2**.

(28) Referring to FIG. **3**, the primary seal device **50** is configured as an annular seal device such as, but not limited to, a non-contact hydrostatic seal device. The primary seal device **50** includes a seal base **62**, a plurality of seal shoes **64**, a plurality of spring elements **66** (see also FIGS. **4** and **5**).

(29) The seal base **62** may be configured as an annular full hoop body. The seal base **62** of FIG. **2** extends circumferentially about (e.g., completely around) the axial centerline **32**. The seal base **62** is configured to extend circumferentially around and thereby circumscribe and support the seal shoes **64** as well as the spring elements **66**. Referring to FIG. **4**, the seal base **62** extends axially along the axial centerline **32** between and forms the first side surface **56** and the second side surface **60**. The seal base **62** extends radially between a seal base inner side **68** and a seal base outer side **70**. The seal base outer side **70** radially engages (e.g., is press fit against or otherwise contacts) the stationary structure **34** and its inner surface **44** as shown in FIG. **1**.

(30) Referring to FIG. **3**, the seal shoes **64** may be configured as arcuate bodies and are arranged circumferentially around the axial centerline **32** in an annular array. Each seal shoe **64**, for example, is arranged circumferentially between and next to a pair of adjacent circumferentially neighboring seal shoes **64**. The annular array of the seal shoes **64** extends circumferentially about (e.g., completely around) the axial centerline **32**, thereby forming an inner bore at an inner side **72** of the primary seal device **50**. As best seen in FIG. **1**, the inner bore is sized to receive the seal land **46**, where the rotating structure **36** projects axially through (or into) the inner bore formed by the seal shoes **64**.

(31) Referring to FIG. **5**, each of the seal shoes **64** extends radially from the inner side **72** of the primary seal device **50** to an outer side **74** of that seal shoe **64**. Each of the seal shoes **64** extends circumferentially about the axial centerline **32** between opposing first and second ends **76** and **78** of that seal shoe **64**. Referring to FIG. **4**, each of the seal shoes **64** extends axially along the axial centerline **32** between a first (e.g., upstream) side **80** and a second (e.g., downstream) side **82** of the seal shoe **64**. The seal shoe first side **80** may be an upstream side relative, for example, to flow of leakage fluid across the primary seal device **50**. The seal shoe first side **80** may be axially offset from and project axially away from the axial first side surface **56**. The seal shoe second side **82** may be a downstream side relative, for example, to the flow of leakage fluid across the primary seal device **50**. The seal shoe second side **82** may be axially aligned with the second side surface **60**. The seal shoes **64** of the present disclosure, however, are not limited to such exemplary relationships.

(32) Each of the seal shoes **64** includes a seal shoe base **84**, one or more seal shoe projections **86A-D** (generally referred to as “**86**”) (e.g., rails/teeth) and at least one aperture **88** (e.g., through hole). The seal shoe base **84** is disposed at (e.g., on, adjacent or proximate) the seal shoe outer side **74**. The seal shoe base **84** of FIG. **5**, for example, includes a (e.g., arcuate) base outer surface **90** at the outer side **74**. Referring to FIG. **4**, the seal shoe base **84** extends radially between the base outer surface **90** and one or more (e.g., arcuate) base inner surfaces **92**. Each of these base inner surfaces

92 may be an arcuate surface. Referring to FIG. 5, the seal shoe base **84** extends circumferentially about the axial centerline **32** between the seal shoe first end **76** and the seal shoe second end **78**. The seal shoe base **84** includes a first end surface at the seal shoe first end **76** and a second end surface at the seal shoe second end **78**. Each of the end surfaces may be a flat planar surface. Each of the end surfaces, for example, may have a straight sectional geometry when viewed, for example, in a plane perpendicular to the axial centerline **32**; e.g., the plane of FIG. 5. Referring to FIG. 4, the seal shoe base **84** extends axially between the seal shoe first side **80** and the seal shoe second side **82**.

(33) The seal shoe base **84** includes a (e.g., arcuate) side surface **94** generally at the seal shoe first side **80**. In the array, these side surfaces **94** collectively form a generally annular, but circumferentially segmented, side surface configured for sealingly engaging with (e.g., contacting) the secondary seal devices **52** as shown in FIG. 1. The seal shoes **64** of the present disclosure, however, are not limited to the foregoing exemplary configuration.

(34) Referring to FIG. 4, the seal shoe projections **86** are arranged at discrete axial locations along the axial centerline **32** and the seal shoe base **84**. Each pair of axially adjacent/neighboring projections **86** may thereby be axially separated by an (e.g., arcuate) inter-projection gap. The seal shoe projections **86** of FIG. 4 are configured parallel to one another.

(35) The seal shoe projections **86** may be arranged in a concentrated grouping **96**. This grouping **96** may be asymmetrically arranged axially along the centerline **32** between the seal shoe first side **80** and the seal shoe second side **82**. For example, an axial center **98** (e.g., midpoint) of the grouping **96** of the seal shoe projections **86** in FIG. 4 is arranged closer to the seal shoe first side **80** than the seal shoe second side **82**. More particularly, the axial center **98** is disposed axially between the seal shoe first side **80** and an axial center **100** (e.g., midpoint) of the respective seal shoe **64** and its seal shoe base **84**. As a result, one or more or each of the seal shoe projections **86** (e.g., projections **86A-C**) may be located axially along the centerline **32** between the seal shoe first side **80** and the axial center **100**. The seal shoe projections **86** of the present disclosure, however, are not limited to the foregoing exemplary asymmetric configuration.

(36) The seal shoe projections **86** are connected to (e.g., formed integral with or otherwise attached to) the seal shoe base **84**. Each of the seal shoe projections **86** projects radially inwards from the seal shoe base **84** and its base inner surfaces **92** to a distal projection end.

(37) Each of the seal shoe projections **86** has a projection inner surface **102A-D** (generally referred to as “**102**”) at the distal projection end. One or more or each of the projection inner surfaces **102** may also be at the inner side **72** of the primary seal device **50**. Each projection inner surface **102** may be an arcuate surface. Each projection inner surface **102**, for example, may have an arcuate sectional geometry when viewed, for example, in a plane perpendicular to the axial centerline **32**; e.g., the plane of FIG. 5. The projection inner surfaces **102** are configured to be arranged in close proximity with (but not touch) and thereby sealingly mate with the outer seal land surface **48** in a non-contact manner (see FIG. 1), where the rotating structure **36** projects axially through (or into) the inner bore formed by the seal shoes **64**.

(38) Each of the seal shoe projections **86** extends axially between opposing projection end surfaces **104**. Each of these end surfaces **104** extends radially between and may be contiguous with a respective one of the projection inner surfaces **102** and a respective one of the base inner surfaces **92**.

(39) Each of the seal shoe projections **86** of FIG. 4 has the same radial height. In other embodiments, however, one or more of the seal shoe projections **86** may have a different radial height than at least another one of the seal shoe projections **86**; e.g., see FIG. 6.

(40) Referring again to FIG. 4, the aperture **88** may be located axially along the centerline **32** between the grouping **96** of the seal shoe projections **86** and the seal shoe second side **82**. The aperture **88** of FIG. 4, for example, is located approximately midway axially between the seal shoe projection **86D** and the seal shoe second side **82**. The aperture **88** may also or alternatively be

located axially between the axial center **100** and the seal shoe second side **82**.

(41) The aperture **88** extends radially relative to the centerline **32** completely through the respective seal shoe **64** and its seal shoe base **84**. The aperture **88** thereby fluidly couples an inner space **106** below the respective seal shoe **64** with an outer space **108** above the respective seal shoe **64**. The aperture **88** may thereby provide the respective seal shoe **64** with an effective axial seal shoe length **110** that is shorter than an actual axial seal shoe length **112** of the respective seal shoe **64** along the centerline **32** between the seal shoe first side **80** and the seal shoe second side **82**.

(42) Referring to FIG. 3, the spring elements **66** are arranged circumferentially about the axial centerline **32** in an annular array. Referring to FIGS. 4 and 5, the spring elements **66** are also arranged (e.g., radially) between the seal shoes **64** and the seal base **62**. Each of the spring elements **66** is configured to moveably and resiliently connect a respective one of the seal shoes **64** to the seal base **62**.

(43) The spring element **66** of FIG. 5 includes first and second mounts **114** and **116** (e.g., radial fingers/projections) and one or more spring beams **118A** and **118B** (generally referred to as “**118**”). The first mount **114** is (e.g., directly) connected to a respective one of the seal shoes **64** at (e.g., on, adjacent or proximate) the seal shoe first end **76**, where the opposing seal shoe second end **78** is free floating (e.g., the seal shoe **64** is cantilevered from the first mount **114**). The first mount **114** projects radially outward from the seal shoe base **84**. The second mount **116** is (e.g., directly) connected to the seal base **62**, and is generally circumferentially aligned with or near the seal shoe second end **78**. The second mount **116** is therefore disposed a circumferential distance from the first mount **114**. The second mount **116** projects radially inward from the seal base **62**.

(44) The spring beams **118** are configured as resilient, biasing members of the primary seal device **50**. The spring beams **118** of FIG. 5, for example, are configured as cantilevered-leaf springs. These spring beams **118** may be radially stacked and spaced apart from one another so as to form a four bar linkage with the first mount **114** and the second mount **116**. More particularly, each of the spring beams **118** is (e.g., directly) connected to the first mount **114** and the second mount **116**. Each of the spring beams **118** extends laterally (e.g., circumferentially or tangentially) between and to the first mount **114** and the second mount **116**. The spring beams **118** of FIG. 5 may thereby laterally overlap a major circumferential portion (e.g., ~65-95%) of the respective seal shoe **64**.

(45) During operation of the primary seal device **50** of FIG. 1, rotation of the rotating structure **36** may develop aerodynamic forces and apply a fluid pressure to the seal shoes **64** causing each seal shoe **64** to respectively move radially relative to the outer seal land surface **48**. The fluid velocity may increase as a gap between a respective seal shoe **64** and the outer seal land surface **48** increases, thus reducing pressure in the gap and drawing the seal shoe **64** radially inwardly toward the outer seal land surface **48**. As the gap closes, the velocity may decrease and the pressure may increase within the gap, thus, forcing the seal shoe **64** radially outwardly from the outer seal land surface **48**. The respective spring element **66** and its spring beams **118** may deflect and move with the seal shoe **64** to enable provision of a primary seal of the gap between the outer seal land surface **48** and seal shoe projections **86** within predetermined design tolerances.

(46) Referring to FIG. 4, the aperture **88** enables fluid (e.g., gas) to flow between the inner space **106** and the outer space **108**. Thus, as the respective seal shoe **64** moves radially inward, some of the fluid within the inner space **106** may move outward into the outer space **108**. Conversely, as the respective seal shoe **64** moves radially outward, some of the fluid within the outer space **108** may move inward into the inner space **106**. The aperture **88** and the resulting reduced effective axial seal shoe length **110** may thereby enable improved damping of the radial movement of the respective seal shoe **64**, particularly relative to a seal shoe configured without a through-hole and, thus, with a longer effective axial seal shoe length; i.e., the actual seal shoe length.

(47) Referring again to FIG. 1, while the primary seal device **50** described above is operable to generally seal the annular gap **40** between the stationary structure **34** and the rotating structure **36**, the fluid (e.g., gas) may still flow axially through passages defined by the radial air gaps **120A-C**

(generally referred to as “120”) between the elements **62**, **118A** and **118B**. The secondary seal devices **52** therefore are provided to seal off these passages and, thereby, further and more completely seal the annular gap **40**.

(48) Each of the secondary seal devices **52** may be configured as a ring seal element such as, but not limited to, a split ring. Alternatively, one or more of the secondary seal devices **52** may be configured as a full hoop body ring, an annular brush seal or any other suitable ring-type seal.

(49) The secondary seal devices **52** of FIG. **1** are arranged together in an axial stack. In this stack, each of the secondary seal devices **52** axially engages (e.g., contacts) another adjacent one of the secondary seal devices **52**. The stack of the secondary seal devices **52** is arranged with the first ring structure **54**, which positions and mounts the secondary seal devices **52** with the stationary structure **34** adjacent the primary seal device **50**. In this arrangement, the stack of the secondary seal devices **52** is operable to axially engage and form a seal between the side surfaces **94** of the seal shoes **64** and an annular surface **122** of the first ring structure **54**. These surfaces **94** and **122** are axially aligned with one another, which enables the stack of the secondary seal devices **52** to slide radially against, but maintain a seal engagement with, the side surfaces **94** as the seal shoes **64** move radially relative to the outer seal land surface **48** as described above.

(50) The first ring structure **54** may include a secondary seal device support ring **124** and a retention ring **126**. The support ring **124** is configured with an annular full hoop body, which extends circumferentially around the axial centerline **32**. The support ring **124** includes the annular surface **122**, and is disposed axially adjacent and engaged with the seal base **62**.

(51) The retention ring **126** is configured with an annular full hoop body, which extends circumferentially around the axial centerline **32**. The retention ring **126** is disposed axially adjacent and is engaged with the support ring **124**, thereby capturing the stack of the secondary seal devices **52** within an annular channel formed between the rings **124** and **126**. The stack of the secondary seal devices **52**, of course, may also or alternatively be attached to one of the rings **124** and **126** by, for example, a press fit connection and/or otherwise.

(52) In some embodiments, referring to FIGS. **7A** and **7B**, the aperture **88** may extend axially within (e.g., not into or through) the respective seal shoe **64** and its seal shoe base **84**. The aperture **88** of FIGS. **7A** and **7B**, for example, extends axially along the centerline **32** within the seal shoe base **84** between an axial aperture first (e.g., upstream) side surface **128** and an axial aperture second (e.g., downstream) side surface **130**, where the aperture second side surface **130** is arranged axially opposite the aperture first side surface **128**.

(53) In some embodiments, referring to FIG. **8**, the aperture **88** may extend axially into the respective seal shoe **64** and its seal shoe base **84**. The aperture **88** of FIG. **8**, for example, extends axially along the centerline **32** into the seal shoe base **84** from the seal shoe second side **82** to the aperture first side surface **128**. The aperture **88** may thereby form a notch or a groove in the respective seal shoe **64** and its seal shoe base **84**.

(54) In some embodiments, referring to FIGS. **7A**, **7B** and **8**, the aperture **88** extends laterally within the respective seal shoe **64** and its seal shoe base **84**. The aperture **88** of FIGS. **7A**, **7B** and **8**, for example, extends laterally (e.g., circumferentially about the centerline **32**) within the seal shoe base **84** between a lateral aperture first side surface **132** and a lateral aperture second side surface **134**, where the aperture second side surface **134** is arranged laterally opposite the aperture first side surface **132**. The aperture first side surface **132** extends axially between and is connected to the side surfaces **128** and **130**. The aperture second side surface **134** extends axially between and is connected to the side surfaces **128** and **130**.

(55) In some embodiments, referring to FIGS. **9** and **10**, the aperture **88** may extend laterally into the respective seal shoe **64** and its seal shoe base **84**. The aperture **88** of FIGS. **9** and **10**, for example, extends laterally into the seal shoe base **84** from the seal shoe first end **76** to the aperture second side surface **134**. The aperture **88** may thereby form a notch or a groove in the respective seal shoe **64** and its seal shoe base **84**.

(56) In some embodiments, referring to FIG. 5, the aperture **88** may be located laterally between the first mount **114** and the second mount **116**.

(57) In some embodiments, referring to FIG. 11, the first mount **114** may be located laterally between the aperture **88** and the second mount **116**.

(58) In some embodiments, referring to FIGS. 12A, 12B, 13A, 13B and 14-17, the aperture **88** may be one of a plurality of apertures **88** configured in the respective seal shoe **64** and its seal shoe base **84**. Each of these apertures **88** may extend radially through the respective seal shoe **64** and its seal shoe base **84**. Some or all of the apertures **88** may be arranged in a (e.g., straight) linear array. The aperture **88**, for example, may be axially aligned with one another along the centerline **32**; however, the present disclosure is not limited to such an exemplary arrangement.

(59) In the embodiments of FIGS. 12A and 13A, each of the apertures **88** is located laterally between the first mount **114** and the second mount **116** (not visible; see FIG. 4). In the embodiments of FIGS. 14-17, the first mount **114** is located laterally between (A) at least one of the apertures **88** and (B) the second mount **116** (not visible; see FIG. 4).

(60) In some embodiments, referring to FIG. 18, one or more or each aperture **88** is configured with an elongated cross-sectional geometry when viewed, for example, in a plane perpendicular to a centerline axis **136** of the aperture **88**; e.g., plane of FIG. 18. This elongated cross-sectional geometry may be laterally elongated as shown in FIG. 18. The aperture **88** of FIG. 18, for example, has a laterally extending major axis **138** and an axially extending minor axis **140**. A length of the major axis **138** may be at least two, five, ten, fifteen, twenty or more times a length of the minor axis **140**. The present disclosure, however, is not limited to the foregoing major-minor axes orientation or dimensional relationship.

(61) In some embodiments, referring to FIG. 19, one or more or each aperture **88** is configured with a non-elongated (e.g., circumferentially symmetrical) cross-sectional geometry when viewed, for example, in a plane perpendicular to the centerline axis **136** of the aperture **88**; e.g., plane of FIG. 19. The aperture **88** of FIG. 19, for example, is configured with a circular cross-sectional geometry. In other embodiments, however, the aperture **88** may be configured with a square cross-sectional geometry or any other shape (e.g., equilateral triangle, star, etc.) cross-sectional geometry that is symmetric about a center point; e.g., the centerline axis **136** of the aperture **88**.

(62) Referring to FIG. 20, each aperture **88** has a lateral length **142** (e.g., minor axis length); e.g., a length laterally between the side surfaces **132** and **134**. This lateral aperture length **142** is sized to be smaller than a lateral seal shoe length **144**; e.g., the length laterally between the seal shoe ends **76** and **78**. In some embodiments, the lateral aperture length **142** may be between about (e.g., +/- 1%) fifty percent (50%) and about eighty percent (80%) of the seal shoe length **144**; e.g., see FIGS. 7A and 7B. In some embodiments, the lateral aperture length **142** may be between about ten percent (10%) and about fifty percent (50%) of the seal shoe length **144**; e.g., see FIGS. 12A and 12B. In some embodiments, the lateral aperture length **142** may be between about one percent (1%) and about ten percent (10%) of the seal shoe length **144**; e.g., see FIGS. 13A and 13B. The present disclosure, however, is not limited to the foregoing exemplary aperture-seal shoe dimensional relationships.

(63) The primary seal device **50** and some or all of its elements (e.g., **62**, **64** and **66**) may be configured as a monolithic body. However, the present disclosure is not limited to such a primary seal device construction.

(64) As described above, the rotational equipment assembly **30** of the present disclosure may be configured with various different types and configurations of rotational equipment. FIG. 21 illustrates one such type and configuration of the rotational equipment—a geared turbofan gas turbine engine **146**. Such a turbine engine includes various stationary structures (e.g., bearing supports, hubs, cases, etc.) as well as various rotors (e.g., rotor disks, shafts, shaft assemblies, etc.) as described below, where the stationary structure **34** and the rotating structure **36** can respectively be configured as anyone of the foregoing structures in the turbine engine **146** of FIG. 21, or other

structures not mentioned herein.

(65) The turbine engine **146** of FIG. **21** extends along the axial centerline **32** between an upstream airflow inlet **148** and a downstream airflow exhaust **150**. The turbine engine **146** includes a fan section **152**, a compressor section **153**, a combustor section **154** and a turbine section **155**. The compressor section **153** includes a low pressure compressor (LPC) section **153A** and a high pressure compressor (HPC) section **153B**. The turbine section **155** includes a high pressure turbine (HPT) section **155A** and a low pressure turbine (LPT) section **155B**.

(66) The engine sections **152-155B** are arranged sequentially along the axial centerline **32** within an engine housing **156**. This engine housing **156** includes an inner case **158** (e.g., a core case) and an outer case **160** (e.g., a fan case). The inner case **158** may house one or more of the engine sections **153A-155B**; e.g., an engine core. The outer case **160** may house at least the fan section **152**.

(67) Each of the engine sections **152**, **153A**, **153B**, **155A** and **155B** includes a respective rotor **162-166**. Each of these rotors **162-166** includes a plurality of rotor blades arranged circumferentially around and connected to one or more respective rotor disks. The rotor blades, for example, may be formed integral with or mechanically fastened, welded, brazed, adhered and/or otherwise attached to the respective rotor disk(s).

(68) The fan rotor **162** is connected to a gear train **168**, for example, through a fan shaft **170**. The gear train **168** and the LPC rotor **163** are connected to and driven by the LPT rotor **166** through a low speed shaft **171**. The HPC rotor **164** is connected to and driven by the HPT rotor **165** through a high speed shaft **172**. The shafts **170-172** are rotatably supported by a plurality of bearings **174**. Each of these bearings **174** is connected to the engine housing **156** by at least one stationary structure such as, for example, an annular support strut.

(69) During operation, air enters the turbine engine **146** through the airflow inlet **148**. This air is directed through the fan section **152** and into a core gas path **176** and a bypass gas path **178**. The core gas path **176** extends sequentially through the engine sections **153A-155B**. The air within the core gas path **176** may be referred to as “core air”. The bypass gas path **178** extends through a bypass duct, which bypasses the engine core. The air within the bypass gas path **178** may be referred to as “bypass air”.

(70) The core air is compressed by the compressor rotors **163** and **164** and directed into a combustion chamber **180** of a combustor in the combustor section **154**. Fuel is injected into the combustion chamber **180** and mixed with the compressed core air to provide a fuel-air mixture. This fuel air mixture is ignited and combustion products thereof flow through and sequentially cause the turbine rotors **165** and **166** to rotate. The rotation of the turbine rotors **165** and **166** respectively drive rotation of the compressor rotors **164** and **163** and, thus, compression of the air received from a core airflow inlet. The rotation of the turbine rotor **166** also drives rotation of the fan rotor **162**, which propels bypass air through and out of the bypass gas path **178**. The propulsion of the bypass air may account for a majority of thrust generated by the turbine engine **146**, e.g., more than seventy-five percent (75%) of engine thrust. The turbine engine **146** of the present disclosure, however, is not limited to the foregoing exemplary thrust ratio.

(71) The assembly **30** may be included in various turbine engines other than the one described above as well as in other types of rotational equipment. The rotational equipment assembly **30**, for example, may be included in a geared turbine engine where a gear train connects one or more shafts to one or more rotors in a fan section, a compressor section and/or any other engine section. Alternatively, the rotational equipment assembly **30** may be included in a turbine engine configured without a gear train. The rotational equipment assembly **30** may be included in a geared or non-geared turbine engine configured with a single spool, with two spools (e.g., see FIG. **21**), or with more than two spools. The turbine engine may be configured as a turbofan engine, a turbojet engine, a propfan engine, a pusher fan engine or any other type of turbine engine. The present disclosure therefore is not limited to any particular types or configurations of turbine engines or

rotational equipment.

(72) While various embodiments of the present disclosure have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the disclosure. For example, the present disclosure as described herein includes several aspects and embodiments that include particular features. Although these features may be described individually, it is within the scope of the present disclosure that some or all of these features may be combined with any one of the aspects and remain within the scope of the disclosure. Accordingly, the present disclosure is not to be restricted except in light of the attached claims and their equivalents.

Claims

1. An assembly for rotational equipment, comprising: a plurality of seal shoes arranged about a centerline in an annular array, the plurality of seal shoes comprising a first seal shoe configured with an aperture that extends radially through the first seal shoe, the first seal shoe extending axially along the centerline between an upstream side of the first seal shoe and a downstream side of the first seal shoe, and the aperture located axially between an axial center of the first seal shoe and the downstream side of the first seal shoe; a seal base circumscribing the annular array of seal shoes; and a plurality of spring elements comprising a first spring element; the first spring element including a first mount, a second mount and a spring beam, the first mount connected to the first seal shoe, the second mount connected to the seal base, and the spring beam extending laterally between and connected to the first mount and the second mount; wherein each of the plurality of seal shoes includes a plurality of projections extending inwardly from the seal base; wherein the plurality of projections in a grouping and are solely between the upstream side and the axial center, and the aperture is solely between the axial center and the downstream side such that the aperture is downstream of the grouping of the plurality of projections; and wherein the aperture is configured with an elongated cross-sectional geometry when viewed in a reference plane perpendicular to a centerline axis of the aperture.
2. The assembly of claim 1, wherein the aperture extends axially within the first seal shoe.
3. The assembly of claim 1, wherein the aperture extends laterally within the first seal shoe.
4. The assembly of claim 1, wherein the aperture extends laterally into the first seal shoe.
5. The assembly of claim 1, wherein the first mount is located laterally between the second mount and the aperture.
6. The assembly of claim 1, wherein the aperture is configured with a circumferentially symmetric cross-sectional geometry.
7. The assembly of claim 1, wherein the aperture is configured with a circular cross-sectional geometry.
8. The assembly of claim 1, wherein the first seal shoe is further configured with a second aperture that extends radially through the first seal shoe.
9. The assembly of claim 1, wherein the seal base extends axially along the centerline between the upstream side of the first seal shoe and the downstream side of the first seal shoe; and the aperture extends radially through the seal base.
10. The assembly of claim 9, wherein the plurality of projections comprises a first projection and a second projection.
11. The assembly of claim 10, wherein the plurality of projections further comprises a third projection.
12. The assembly of claim 1, wherein the first seal shoe extends laterally between a first end of the first seal shoe and a second end of the first seal shoe for a seal shoe length; and the aperture extends laterally for an aperture length which is between fifty percent and eighty percent of the seal shoe length.

13. The assembly of claim 1, wherein the first seal shoe extends laterally between a first end of the first seal shoe and a second end of the first seal shoe for a seal shoe length; and the aperture extends laterally for an aperture length which is between ten percent and fifty percent of the seal shoe length.

14. The assembly of claim 1, wherein the first seal shoe extends laterally between a first end of the first seal shoe and a second end of the first seal shoe for a seal shoe length; and the aperture extends laterally for an aperture length which is between one percent and ten percent of the seal shoe length.

15. The assembly of claim 1, wherein the first spring element further includes a second spring beam that extends laterally between and is connected to the first mount and the second mount.

16. The assembly of claim 1, further comprising: a stationary structure; a rotating structure configured to rotate about the centerline; and a seal assembly including the plurality of seal shoes, the seal base and the plurality of spring elements, the seal assembly configured to seal a gap between the stationary structure and the rotating structure; wherein the plurality of seal shoes are arranged circumferentially about and sealingly engage the rotating structure; and wherein the seal base is mounted to the stationary structure.

17. The assembly of claim 1, further comprising: a seal assembly configured to seal a gap between the seal base and the first seal shoe; the seal assembly comprising a seal device configured to axially engage a side surface of the first seal shoe; and the side surface of the first seal shoe located axially between the upstream side of the first seal shoe and the axial center of the first seal shoe.
